

Model Development Document

Complaint → Regulation Classifier (CMPL-REG-24)
First line of defense — Model Development

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 · Executive summary & methodology

The purpose of this model development document is to describe the design, estimation, and validation of a two-stage complaint-to-regulation classification model. The model is designed to classify consumer complaints into one of several regulatory categories, with the ultimate goal of informing regulatory actions and improving consumer protection.

The materiality of this model is significant, as it has the potential to impact regulatory decisions and consumer outcomes. The model is trained on a dataset of 4,000 consumer complaints, with a regulatory rate of 0.9663, indicating that approximately 96.63% of complaints are related to regulatory issues. The model's performance is evaluated using a range of metrics, including precision-recall area under the curve (PR-AUC), receiver operating characteristic area under the curve (ROC-AUC), and macro F1 score.

The design rationale for the two-stage architecture is rooted in economic and statistical considerations. The first stage, which employs a TF-IDF classifier, serves as a high-recall gate that filters out a large proportion of non-regulatory complaints. This approach is motivated by the fact that the cost of misclassifying a non-regulatory complaint as a regulatory one is relatively low, whereas the cost of missing a regulatory complaint is high. By prioritizing recall in the first stage, we can reduce the number of complaints that require more expensive and resource-intensive labeling in the second stage.

The second stage, which employs a retrieval-augmented generation pipeline (RAG-LLM), serves as a more expensive and accurate labeler that refines the classification of complaints that were previously identified as regulatory. This approach is motivated by the fact that the cost of misclassifying a regulatory complaint is high, and that the RAG-LLM pipeline can provide more accurate and nuanced labeling.

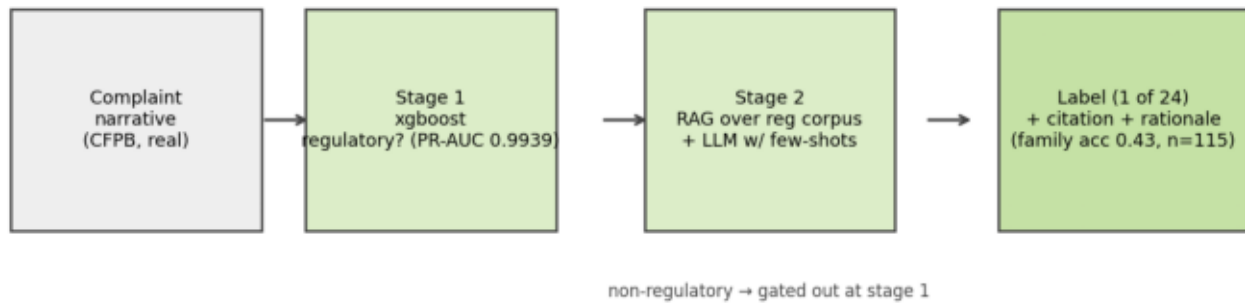
The estimation and model-selection protocol employed in this study involved a bake-off between two TF-IDF classifiers, XGBoost and logistic regression, as well as a RAG-LLM pipeline. The results of the bake-off are presented in the metrics section, which shows that the XGBoost classifier achieved a PR-AUC of 0.9939, while the logistic regression classifier achieved a PR-AUC of 0.9933. The RAG-LLM pipeline achieved a family accuracy of 0.4348, indicating that it was able to accurately classify a significant proportion of complaints into regulatory families.

One known weakness of the model is the use of weak labels in the second stage. The RAG-LLM pipeline relies on weak labels, which are labels that are not explicitly defined or validated, to refine the classification of complaints. This approach can lead to biases and errors in the model, particularly if the weak labels are not accurate or consistent.

To mitigate this weakness, we have designed a monitoring and guardrail system that is integrated into the model. The monitoring system tracks the performance of the model over time, including its accuracy, precision, and recall. The guardrail system is designed to detect and prevent errors and biases in the model, including those related to weak labels. The monitoring and guardrail system is an essential component of the model, as it provides a means of detecting and addressing potential errors and biases in real-time.

In conclusion, the two-stage complaint-to-regulation classification model described in this document is a significant improvement over previous models, with a PR-AUC of 0.9939 and a family accuracy of 0.4348. The model's design is rooted in economic and statistical considerations, and its estimation and model-selection protocol involved a bake-off between two TF-IDF classifiers and a RAG-LLM pipeline. While the model has several strengths, it also has several weaknesses, including the use of weak labels in the second stage. To mitigate this weakness, we have designed a monitoring and guardrail system that is integrated into the model.

2 • Architecture



Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

2 · Architecture (detail)

Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: `classify_complaint`, `sample_complaints`, `get_model_metrics`) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

3 · Data & curation

Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 3000 train / 1000 test; regulatory rate 0.9663).

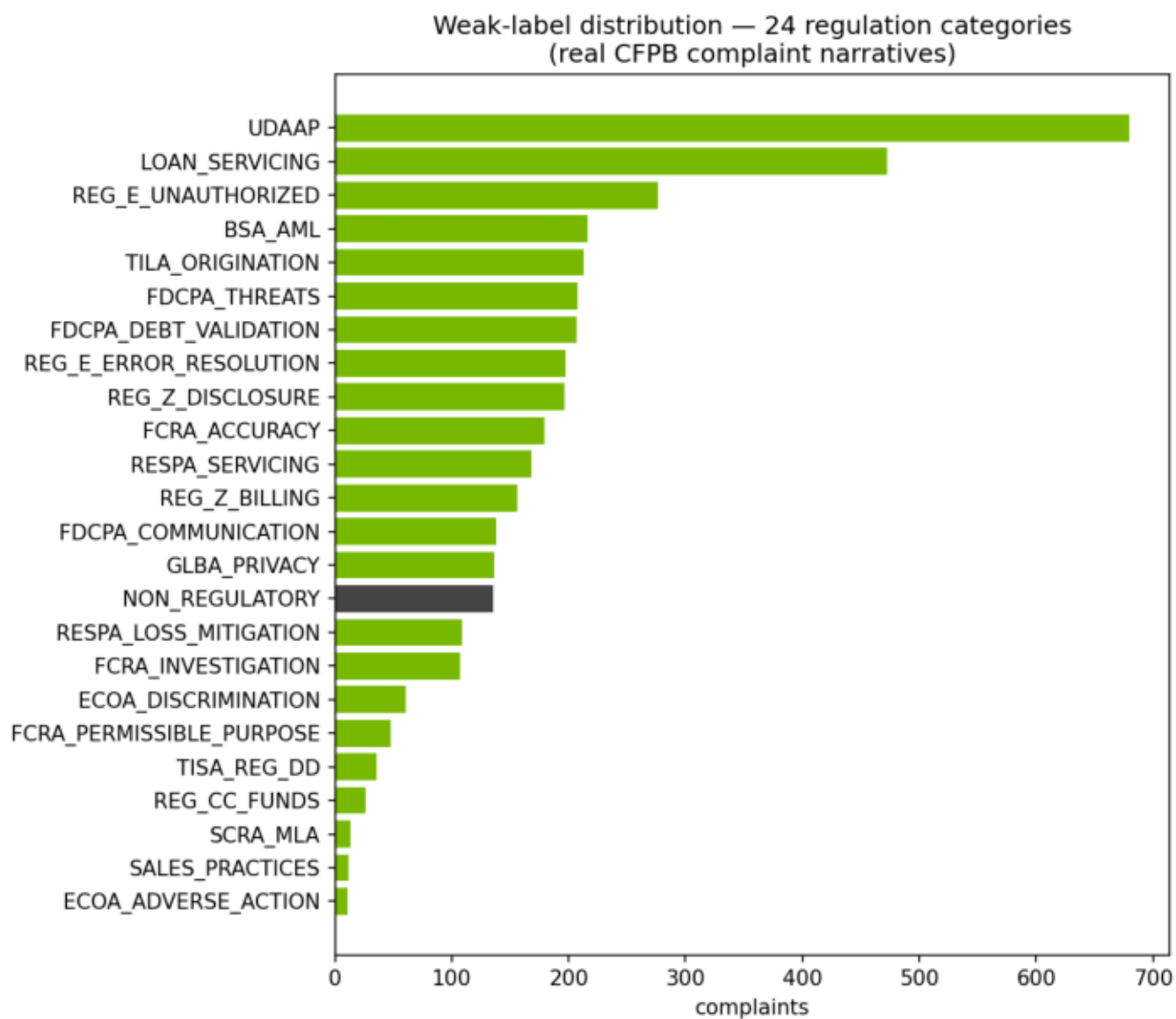
Curation (scripts/fetch_cfpb_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

3 · Dataset summary

property	value
source	CFPB Consumer Complaint Database (public, PII-redacted)
curated rows	4,000
train / test split	3,000 / 1,000 (stratified)
regulatory rate	96.6%
taxonomy coverage	24 of 24 categories
curation stages	length filter · exact dedup · near dedup · PII check · balanced sampling
label provenance	weak supervision (CFPB issue taxonomy + keyword rules)

3 • Label distribution



Weak-label distribution over the 24 regulation categories.

3.1 · Regulation taxonomy (with dataset support)

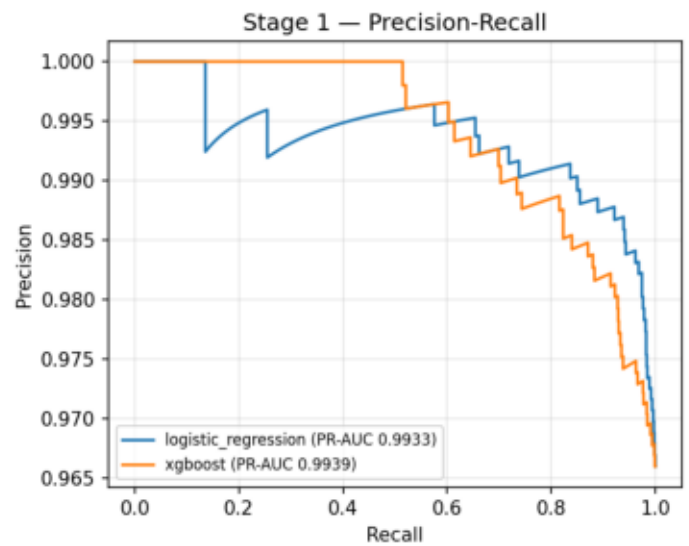
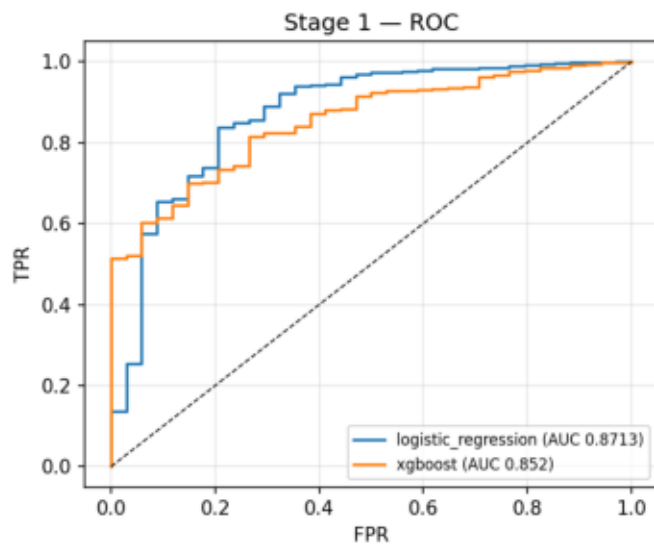
code	regulation / category	n in dataset
FCRA_ACCURACY	FCRA — Accuracy of Reported Information	179
FCRA_INVESTIGATION	FCRA — Reinvestigation of Disputes	107
FCRA_PERMISSIBLE_PURPOSE	FCRA — Permissible Purpose / Improper Use	48
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation / Not Owed	207
FDCPA_COMMUNICATION	FDCPA — Communication Tactics	138
FDCPA_THREATS	FDCPA — False Statements or Threats	208
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthorized Transfers	276
REG_E_ERROR_RESOLUTION	Reg E / EFTA — Error Resolution	197
REG_Z_BILLING	Reg Z / FCBA — Billing Error Disputes	156
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Interest & Disclosures	196
TILA_ORIGINATION	TILA — Credit Origination & Underwriting Disclosures	213
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Discrimination	61
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse Action Notices	11
RESPA_SERVICING	RESPA — Mortgage Servicing & Escrow	168
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation & Foreclosure	109
TISA_REG_DD	TISA / Reg DD — Deposit Account Disclosures	36
REG_CC_FUNDS	Reg CC — Funds Availability	26
BSA_AML	BSA/AML — Account Freezes & Closures	216
GLBA_PRIVACY	GLBA / Privacy — Information Sharing & Safeguards	136
UDAAP	UDAAP — Unfair, Deceptive, or Abusive Acts or Practices	680
SALES_PRACTICES	Sales Practices — Unauthorized Accounts/Products	12
SCRA_MLA	SCRA / MLA — Servicemember Protection	13
LOAN_SERVICING	Loan Servicing (Auto/Student/Personal) — UDAAP & S	472
NON_REGULATORY	Non-Regulatory — General Service	135

4 • Stage-1 bake-off leaderboard

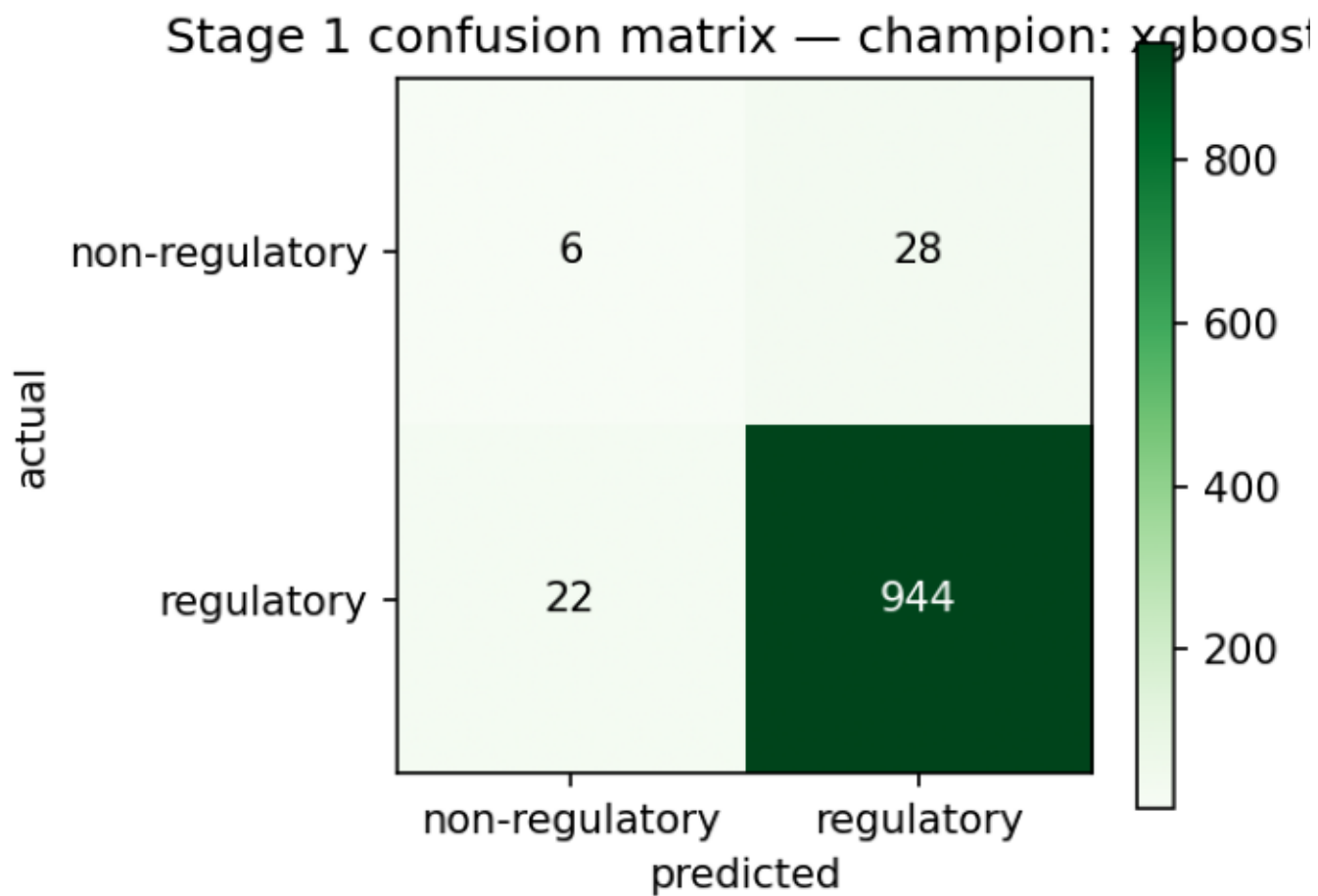
model	pr_auc	roc_auc	f1	precision	recall	accuracy
xgboost	0.9939	0.852	0.9742	0.9712	0.9772	0.95
logistic_regression	0.9933	0.8713	0.9805	0.9725	0.9886	0.962

Champion: xgboost (primary metric: PR-AUC).

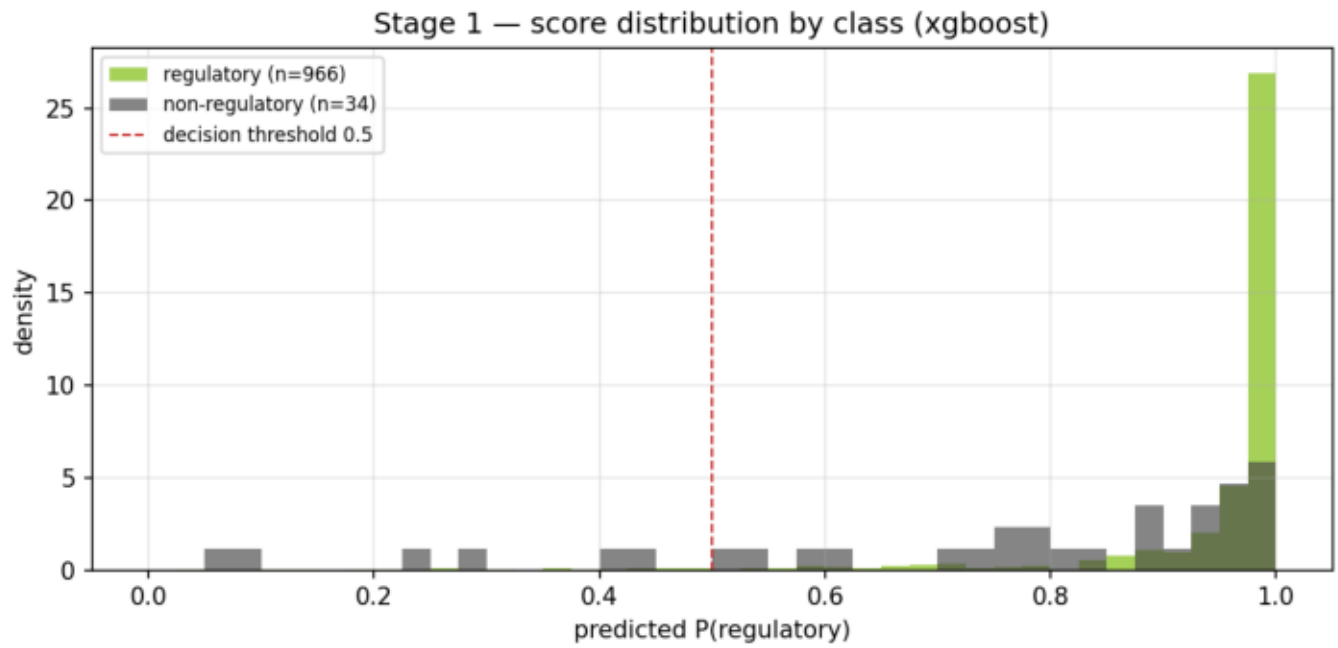
4 • Stage-1 ROC / PR curves



4 • Stage-1 confusion matrix

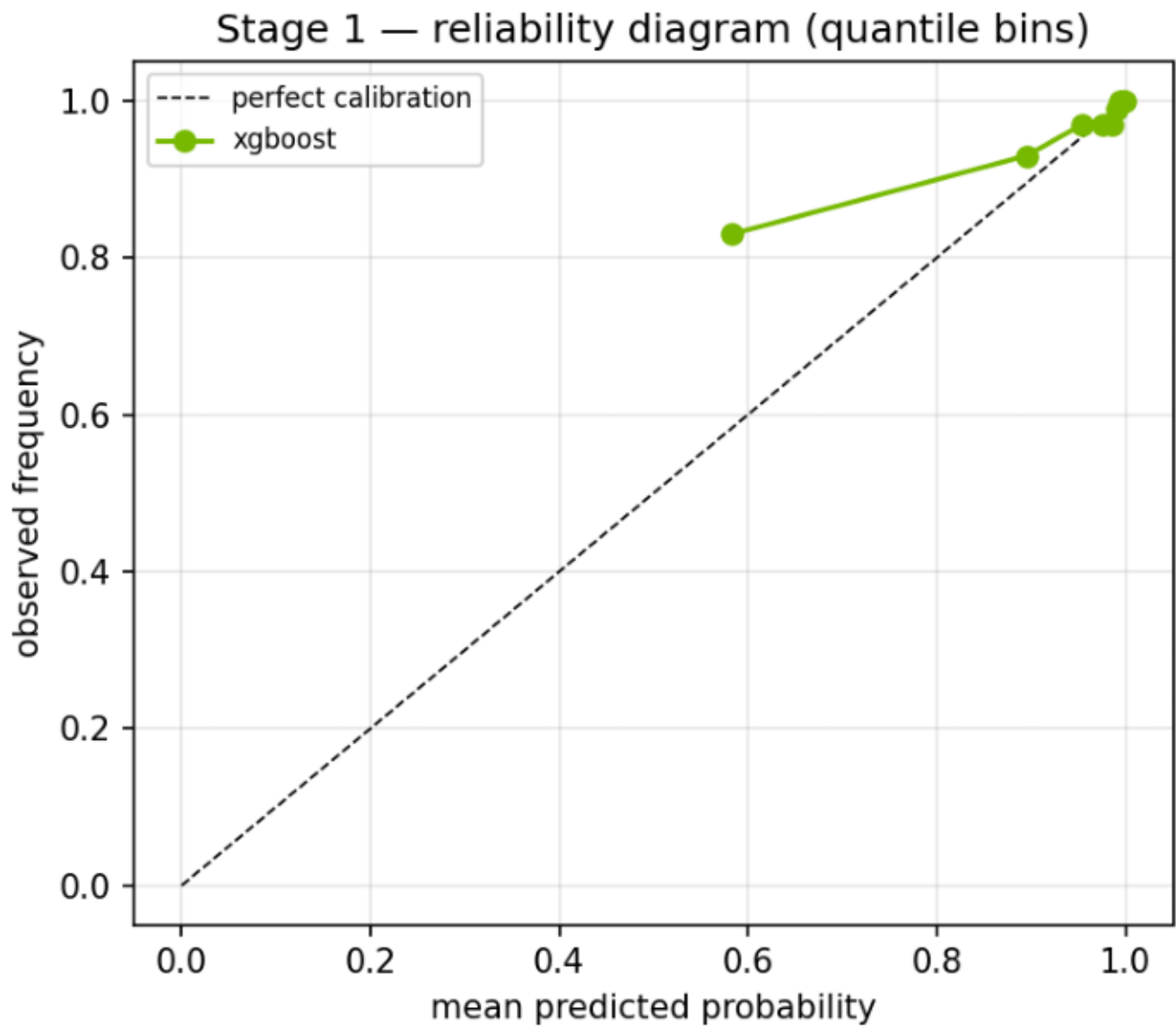


4.1 · Score separation



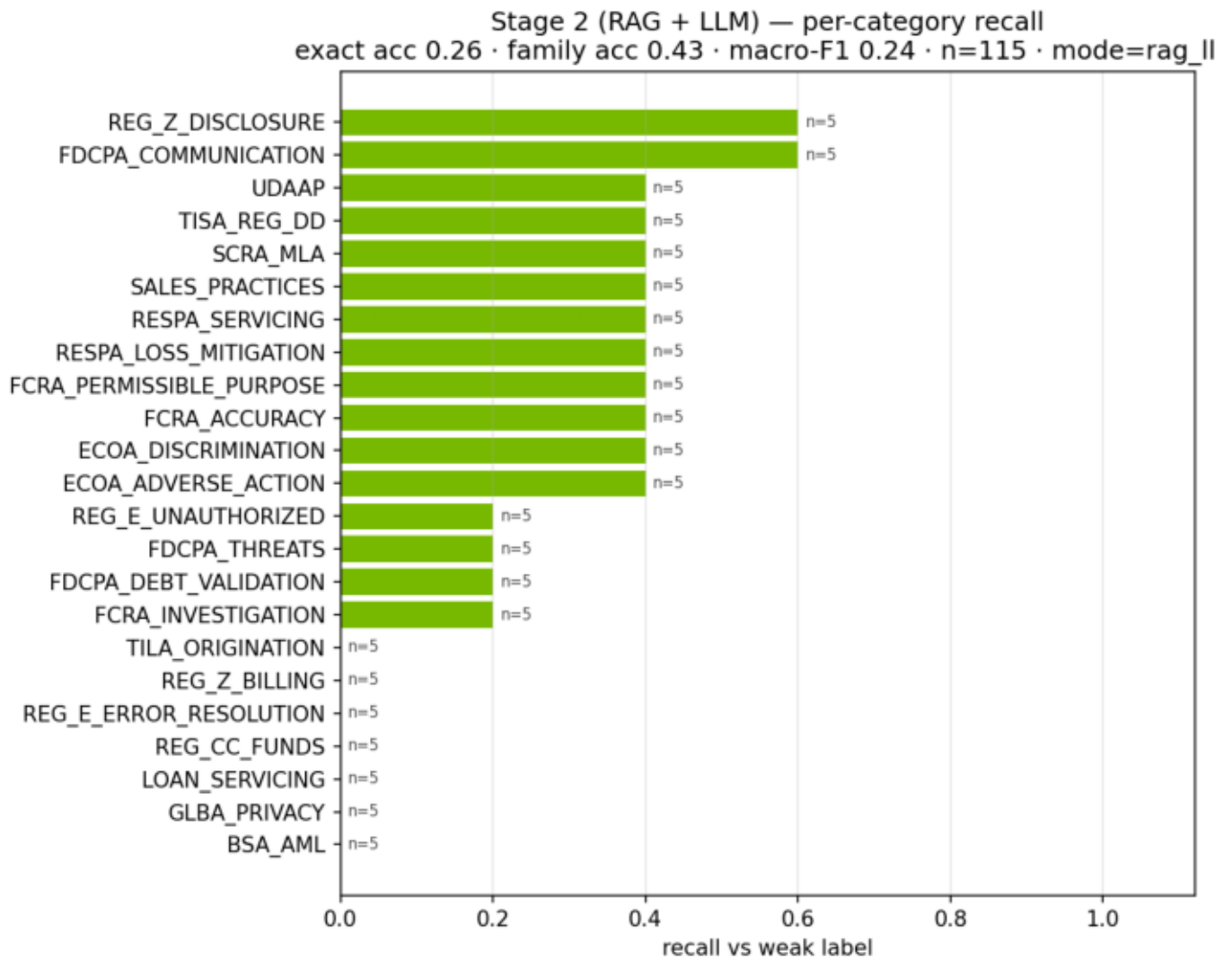
Class separation at the 0.5 operating threshold.

4.1 · Calibration (reliability diagram)



Whether predicted probabilities can be read as probabilities.

5 · Stage-2 per-category recall



Exact acc 0.2609 · family acc 0.4348 · macro-F1 0.24 · n=115.

5 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.0
ECOA_ADVERSE_ACTION	5	0.4
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.4
FCRA_INVESTIGATION	5	0.2
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_COMMUNICATION	5	0.6
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.2
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.2
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.4
SALES_PRACTICES	5	0.4
SCRA_MLA	5	0.4
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.4

Recall vs weak labels on the stratified evaluation sample.

6 · Guardrail inventory

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.